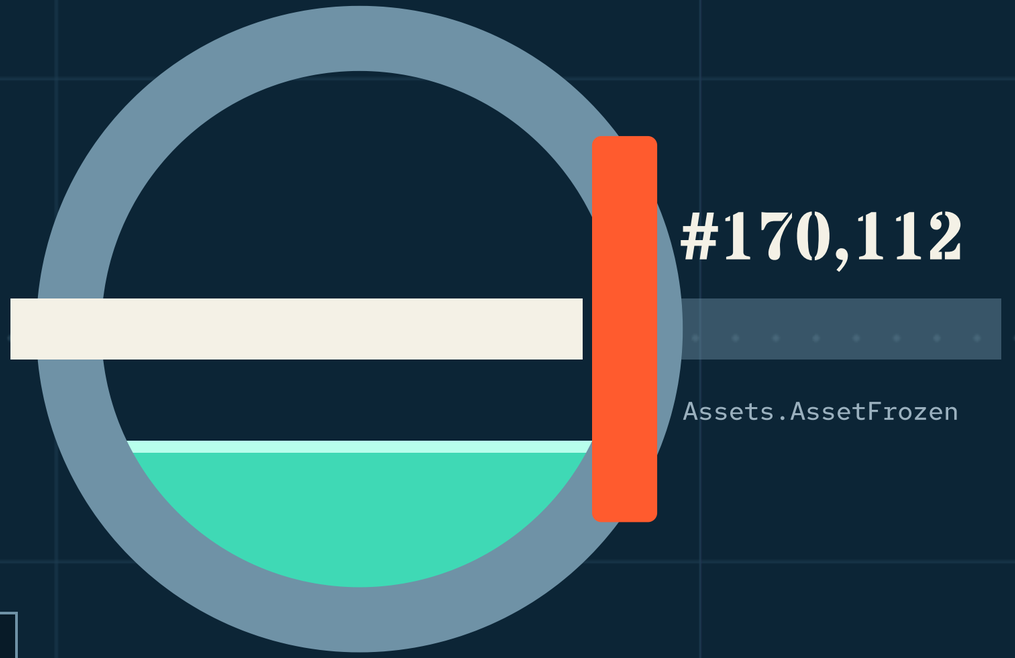
PORTALDOT HACKER HOUSE 2026 · MULTI-CHAIN & CROSS-CHAIN INFRA

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WHEN THE VAULT RUNS SHORT, THE TEST DOLLAR STOPS MOVING.

BALLAST a reserve-gated circuit breaker for a custodial test dollar on Portaldot.



re-run read-only 2026-10-25 · evidence/check-receipts-v3-20261025T013504Z.txt

```
$ python scripts/check_receipts.py --profile v3
[v3] 79/79 receipts confirmed from 8 run(s)
[v3] all confirmed rows at or below the finalized head #182414
```

On the Portaldot V3 testnet, 2026-10-24: watcher W2 froze asset 4401 in extrinsic `170112-1`.
Reserve: local anvil chain, NOT Sepolia. Deck 2026-10-25 · `v0.0.1`

THE PROBLEM

RESERVES DRAIN. NOBODY SEES.

As far as we found, Portaldot has no stable unit of account. The realistic first step is a custodial wrapped dollar, and when its reserves drain, every holder keeps trading an IOU that is already worth less than its face value.

01

104.00

in supply: 54.00 pUSDC-test (asset 4401) + 50.00 control asset (4402), matched 1:1 by test tokens in the vault.

02

−31.20

The custodian calls `release()`: 30% of the reserve leaves the vault.

03

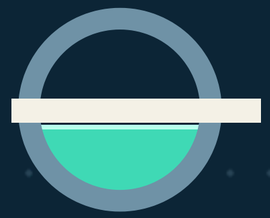
72.80

Reserves left. Supply still says 104.00 unless something on Portaldot acts.

Figures from Ballast's own V3 drill (`evidence/roles-drill-v3-20261024T045513Z.json`; reserve on a local anvil chain, NOT Sepolia). The person it protects: a dApp builder deciding whether checkout may accept pUSDC-test.

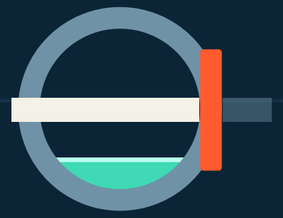
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FOUR ROLES, THREE MULTISIGS



LEVEL

reserves ≥ supply · asset Live · transfers work



TRIPPED

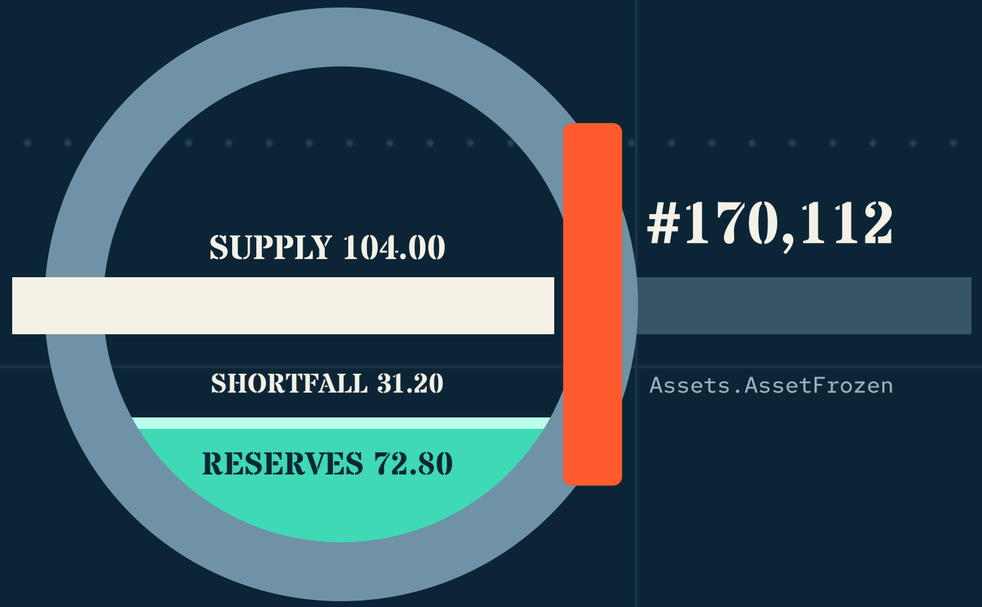
reserves < supply · asset Frozen · `Assets.transfer` refused

ROLE	HELD BY	WHAT IT MAY DO
FREEZER	`M_watch` 1-of-2 watchers W1, W2	Any one watcher freezes the asset when reserves < supply. Its thaw attempts fail with `Assets.NoPermission`.
ISSUER	`M_mint` 2-of-2 operator A, reserve bot B	The bot co-signs a mint only if *finalized* reserves cover supply plus the amount, and never while frozen.
ADMIN	`M_admin` 2-of-3 A, A2, B	The bot co-signs a thaw only after finality passes the drain block and reserves cover supply again.
RECORD	bot B remarks + EVM mirror	`BALLAST/1` attestations on chain, mirrored into `ReserveOracle`; `ballast verify` re-reads both chains.

Roles are set with `Assets.set_team`; the reserve check itself runs off chain, in the bot and the watchers.

RESERVES SANK. ASSET FROZE.

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anvil #1622
RESERVE CHAIN

Custodian drains 30% (31.20) with `release()`.

170112-1
V3 · WATCHER W2

`M_watch` freezes asset 4401 (`Assets.freeze_asset`, Ok). The reason rides in the same extrinsic:

```
BALLAST/1|freeze|a=4401|reason=deficit|s=104000000|r=72800000|p=anvil|eb=1622|eh=0x3987686a…
```

170114-1
V3 · A HOLDER

`Assets.transfer` 1.00 → `Err(Assets.AssetNotLive)`

170115-1
V3 · WATCHER W1

A watcher tries `thaw_asset` → `Err(Assets.NoPermission)`: the freezer cannot undo its own freeze.

Drain to freeze included: 12.22 s (wall clock, one machine). Gauge = the freeze remark: s=104000000 (supply of asset 4401 + control asset 4402), r=72800000, 6 decimals. Reserve: local anvil chain, NOT Sepolia.

LIVE PROOF · THE FULL ROLE DRILL ON V3

EVERY GATE, SIGNED OR REFUSED

evidence/roles-drill-v3-20261024T045513Z.json · bot B decisions (… = cut) · reserve: local anvil, NOT Sepolia

```
$ ballast bot cosign-mint 1.00 …  signed
  finalized reserves 104000000 (latest 104000000) >= S1 53000000 + S2 50000000 + amount 1000000 =
    104000000
$ ballast bot cosign-mint 0.01 …  refused
  over-mint: finalized reserves 104000000 < S1 54000000 + S2 50000000 + amount 10000 = 104010000
$ ballast bot cosign-mint 0.01 …  refused   (asset frozen)
  I7: the breaker asset is frozen — no mint over a tripped breaker
$ ballast bot cosign-thaw  refused
  finalized block #1559 is older than the deficit block #1622: finality has not reached the drain yet
$ ballast bot cosign-thaw  signed
  finalized reserves 104000000 … >= S1 54000000 + S2 50000000 = 104000000, finalized block #1624 past
    the freeze
$ ballast bot cosign-team …  refused   the freezer would change …
$ ballast bot cosign-team …  refused   admin must stay M_admin … another admin could thaw without the
  bot …
```

ON CHAIN, V3 TESTNET

170098-1	mint 1.00 executes
170112-1	freeze, asset Frozen
170114-1	transfer refused: `AssetNotLive`
170115-1	watcher thaw refused: `NoPermission`
170133-1	gated thaw, asset Live
170134-1	transfer 1.00 Ok
170156-1	safe `set_team`, co-signed

79/79
V3 extrinsics in 8 runs (4 ERROR runs kept as history), all final ≤ #170291

22/22
role drill steps, 20/20 state checks

15/15
attestation drill steps, 6/6 checks

16/16
role setup steps, 6/6 checks

ARCHITECTURE

THE RUNTIME ENFORCES THE ROLES

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PORTALDOT V3 RUNTIME · ENFORCEMENT

solid: Portaldot V3 runtime · dashed: off-chain processes · bottom: the reserve chain

Assets · asset 4401 pUSDC-test (+ control asset 4402, keyless roles)
owner = issuer = `M_mint` · admin = `M_admin` · freezer = `M_watch`

System remarks
`BALLAST/1` attest · heartbeat · freeze

Revive · ReserveOracle
`post()` attester only · `status()` · `isBacked(maxAge)` for EVM dApps
`0x4832…155e`

`freeze_asset` | `mint · set_team` | `thaw_asset`

M_watch · 1-of-2
`as_multi_threshold_1`

M_mint · 2-of-2
A, B · `as_multi`

M_admin · 2-of-3
A, A2, B · `as_multi`

`batch_all[remark, freeze]` | `A: first approval` | `B: final, if the gate passes` | `attest · heartbeat` | `reads`

watchdog
keys W1, W2 · freeze only · reads latest on each provider

operator
keys A, A2 · proposes mint, thaw, team

reserve bot
key B · co-signs mint, thaw, set_team · SQLite decision log

attester
key B · remark, then `post()` to the oracle

ballast verify
re-reads both chains; rejects wrong signers

`balanceOf @ latest, each provider` | `@ finalized and latest` | `@ finalized` | `@ block hash`

Reserve chain · ReserveVault.sol + ERC-20 reserve token
`reserves()` · `release()` owner-only · in these drills: a local anvil chain, NOT Sepolia · target: Sepolia with Circle test USDC

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THE STOP LIVES IN THE ASSET

QUESTION	A RESERVE NUMBER ALONE (FRAMING, NOT MEASURED)	BALLAST, MEASURED ON V3
WHO STOPS TRANSFERS?	each dApp that chose to read it	the runtime refuses `Assets.transfer` → `AssetNotLive` (`170114-1`; other transfer paths not drilled)
CAN THE STOPPER RESTART IT?	the number carries no role on the asset	no: a watcher's thaw → `NoPermission` (`170115-1`)
WHEN CAN IT RESTART?	whenever a consumer decides	the bot co-signs a thaw only once finalized reserves pass the drain: refused at anvil #1559 < #1622, co-signed in `170133-1`. A and A2 together could still thaw without it.
CAN SOMEONE FAKE THE RECORD?	depends on who signs the feed	a forged remark from another key (`170253-1`) → `ballast verify` REJECTED; 4/4 bot attestations PASS

Built from pallet-native modules (Assets roles, Multisig, System remarks) with no contract code in the enforcement path. The one contract, `ReserveOracle`, is a read-only mirror: it cannot freeze or thaw anything.

ECOSYSTEM FIT

PORTALDOT'S OWN MODULES

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ASSETS

`create` · `set_metadata` · `set_team` · `mint` · `freeze_asset` · `thaw_asset` · `transfer`

breaker asset 4401 and control asset 4402 on V3

MULTISIG

`as_multi` · `as_multi_threshold_1` · `cancel_as_multi`

three role multisigs: 2-of-2, 2-of-3, 1-of-2

UTILITY

`batch_all`

the freeze and its reason remark land as one extrinsic

SYSTEM

`remark_with_event`

`BALLAST/1` attestations, heartbeats, freeze reasons

REVIVE

`instantiate_with_code` · `call`

`ReserveOracle` on V3 (deployed in 169960-1) exposes `isBacked(maxAge)` to EVM dApps; called on the stand-in, not yet on V3

PINNED TO THE LIVE CHAIN

GENESIS

`0xd4f9acbc…5283bd`

RUNTIME

`portaldot-v3 / 3002`

BLOCKS

`6000 ms`

OPEN SOURCE

MIT · one CLI, `ballast` · 274 tests (239 pytest, 20 Foundry, 15 Hardhat)

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WHO NEEDS A LOAD LINE

The first issuer of a dollar on Portaldot, who needs a safety layer it can show, and that the ecosystem can require of any issuer.

The dApp builder deciding whether checkout may accept that dollar, who must tell users why it is safe.

PATH · STEPS 2 AND 3 ARE PLANS

1

TESTNET, TODAY

V3 drills pass; one person holds every key; reserve on local anvil.

2

SEPOLIA RESERVE

`ReserveVault` with 100.00 test USDC; a watcher on separate infrastructure.

3

A REAL ISSUER

watcher and attester keys held by parties other than the issuer; community-watcher mode.

Funding (plan): an ecosystem grant to reach a Sepolia-backed deployment with outside watchers; after that, the dollar's issuer runs Ballast as the safety layer the ecosystem could ask of it. **Today:** no external users and no revenue.

STATUS VS. THE FINAL

DONE ON V3. NEXT: THE FINAL

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DONE

- Role split + `ReserveOracle` on V3: PASS `16/16 + 6/6`
- Role drill on V3: PASS `22/22 + 20/20`
- Attestation drill on V3: PASS `15/15 + 6/6`
- `79/79` V3 extrinsics (8 runs) final, re-queried by `check_receipts.py --profile v3`
- `274` tests · `130` commits from 2026-10-14 to build `0071879`

NEXT

- Sepolia `ReserveVault` with 100.00 test USDC (the wallet holds 80.00), then the drill with the real finality wait
- Outage and heartbeat drills on V3 (stand-in only so far)
- A watcher on separate infrastructure
- Watcher-set rotation, `M_admin` rotation, wind-down
- Benchmarks over many drills; a reserves page

Scan for the repository.

REPOSITORY
`github.com/edycutjong/portaldot-hackathon-2026-ballast-edycu`

V3 PROOF TABLES
`docs/PROOF.md`

THE FREEZE, IN THE NODE CONSOLE
`block #170,112`

AI assistance: built with Claude Code (Anthropic). Most of the code, the tests and the prose were drafted with AI assistance and reviewed by the author. The architecture, the role map and the on-chain design decisions are the author's.